Cybersecurity Incident Report

Project: TCP SYN Flood DoS Attack – Network Incident Analysis

Context: Simulated case study | Google Cybersecurity Professional Certificate

Scope: Analysis of provided TCP/HTTP traffic records; no live network investigation was performed.

Section 1 — Identify the Type of Attack

One potential explanation for the website's connection timeout error message is:

The web server was overwhelmed by an unusually high volume of TCP SYN requests, reducing its ability to respond to legitimate connections and causing timeouts.

The logs show that:

The apparent source IP address 203.0.113.0 repeatedly sent TCP SYN packets to the web server at 192.0.2.1 on destination port 443. The repeated requests coincide with connection resets and an HTTP 504 Gateway Time-out affecting legitimate visitors.

This event could be:

A Denial-of-Service (DoS) attack, specifically a TCP SYN flood. The records show one apparent suspicious source, so they do not establish a distributed DDoS attack. Source IP spoofing cannot be ruled out from these records alone.

Section 2 — Explain How the Attack Affects the Website

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake:

1. SYN: The client sends a TCP SYN packet to request a connection.

2. SYN-ACK: The server acknowledges the request and sends a SYN-ACK response.

3. ACK: The client acknowledges the server response, completing the handshake and allowing application data to be exchanged.

Explain what happens when a malicious actor sends a large number of SYN packets all at once:

In a SYN flood, excessive connection requests consume the server's capacity to manage incoming connections. Many requests may remain incomplete or be repeated, exhausting connection-handling resources. Legitimate visitors can experience slow responses, rejected connections, or timeouts.

Explain what the logs indicate and how that affects the server:

Normal traffic initially completes the TCP handshake and receives HTTP 200 OK (packets 47–51). Repeated SYN requests from 203.0.113.0 appear in packets 57, 59, 61, and continue thereafter. Packet 73 shows a TCP RST-ACK sent to a legitimate visitor, while packet 77 records HTTP 504 Gateway Time-out. These observations indicate that service availability degraded as the suspicious traffic persisted.

Evidence Summary

Packet(s)	Observed event	Meaning
47–51	TCP handshake; HTTP 200 OK	Normal successful request
57, 59, 61	Repeated SYN from 203.0.113.0	Suspicious request pattern
73	RST, ACK to 198.51.100.16	Legitimate connection reset
77	HTTP 504 to 198.51.100.5	Gateway timeout
78	Further SYN from 203.0.113.0	Suspicious activity persists

Impact and Recommendations

The incident disrupted employees' access to the sales webpage, potentially delaying customer support and business operations. To reduce recurrence, the organization should consider SYN flood protections such as SYN cookies, rate limiting, connection monitoring, firewall filtering, and upstream DDoS mitigation. Blocking one IP address is a temporary measure and may not stop spoofed or multi-source traffic.

Source and Limitations

Based on the fictional scenario and supplied Wireshark TCP/HTTP log from the Google Cybersecurity Professional Certificate. The log excerpt supports a suspected SYN flood and observed availability failures; it does not independently establish the attacker's identity or intent.